IGCSE 0607 Practice Questions — E1.17 — Combined Questions & Markscheme

Surds

Total Questions: 33 — **Total Marks:** 161 Marks

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- **E1.17** Surds
  → **Paper 2 (Non-Calculator)** *(33 Qs)*

1 Simplify

$$\sqrt{75} - \sqrt{27} + \sqrt{12}$$

..... **[2]**

2 Simplify.

$\sqrt{75} + \sqrt{48} - \sqrt{27}$

..... **[2]**

3 Simplify fully.

$\sqrt{72} + \sqrt{50} - \sqrt{18}$

..... **[2]**

4 Simplify

$$\sqrt{75} - \sqrt{27} + \sqrt{12}$$

..... **[2]**

5 Simplify.

$\sqrt{75} + \sqrt{48} - \sqrt{27}$

..... **[2]**

6 **(a)** Simplify.

$\sqrt{45}$

..... **[1]**

(b) Simplify.

$\sqrt{20} + \sqrt{80}$

..... **[2]**

7 Simplify

$$\sqrt{75} + \sqrt{27} - \sqrt{12}$$

..... **[3]**

8 Simplify the following expression, leaving your answer in the form $k\sqrt{3}$.

$\sqrt{48} - \sqrt{12}$

..... **[3]**

9 **(a)** Simplify.

$\sqrt{75}$

..... **[2]**

(b) Simplify.

$2\sqrt{18} - \sqrt{8}$

..... **[2]**

10 (a) (i) Simplify.

$$\sqrt{45} + \sqrt{20}$$

..... **[2]**

(ii) Rationalise the denominator.

$$\frac{18}{\sqrt{3}}$$

..... **[2]**

11 **(a)** Simplify.

$\sqrt{45}$

..... **[1]**

(b) Work out the value of the following expression, giving your answer in the form $k\sqrt{2}$.

$\sqrt{72} + \sqrt{50} - \sqrt{18}$

..... **[2]**

(c) Rationalise the denominator.

$\dfrac{21}{\sqrt{7}}$

..... **[2]**

12 **(a)** Expand and simplify.

$$(4 - \sqrt{3})(2 + \sqrt{3})$$

..... **[2]**

(b) Rationalise the denominator and simplify.

$$\frac{21}{\sqrt{7}}$$

..... **[2]**

13 **(a)** Expand and simplify

$$(3 - \sqrt{5})^2$$

..... **[2]**

(b) Rationalise the denominator.

$$\frac{14}{\sqrt{7}}$$

..... **[2]**

14 **(a)** Expand and simplify.

$$(3 - 2\sqrt{5})(4 + \sqrt{5})$$

..... **[2]**

(b) Rationalise the denominator and simplify.

$$\frac{14}{\sqrt{7}}$$

..... **[2]**

15 (a) **(a)** Expand and simplify

$$(3 - 2\sqrt{5})(4 + \sqrt{5})$$

..... **[2]**

(b) Rationalise the denominator.

$$\frac{15}{\sqrt{3}}$$

..... **[2]**

16 **(a)** Expand and simplify

$$(3+\sqrt{2})(4-\sqrt{2})$$

..... **[2]**

(b) Rationalise the denominator.

$$\frac{22}{5-\sqrt{3}}$$

..... **[3]**

17 **(a)** Expand and simplify.

$$(2\sqrt{5}-3)(2\sqrt{5}+3)$$

..... **[2]**

(b) Rationalise the denominator.

$$\frac{14}{3-\sqrt{2}}$$

..... **[3]**

18 **(a)** Expand and simplify.

$$(3\sqrt{2}-1)(2\sqrt{2}+5)$$

..... **[2]**

(b) Rationalise the denominator and simplify.

$$\frac{26}{4-\sqrt{3}}$$

..... **[3]**

19 **(a)** Rationalise the denominator and simplify fully.

$$\frac{15}{\sqrt{5}}$$

..... **[2]**

(b) Expand and simplify.

$$(2+\sqrt{3})(5-\sqrt{3})$$

..... **[3]**

20 **(a)** Expand and simplify.

$$(3+\sqrt{5})(2-\sqrt{5})$$

..... **[2]**

(b) Rationalise the denominator and simplify.

$$\frac{14}{\sqrt{7}}$$

..... **[3]**

21 **(a)** Expand and simplify.

$$(5 - 2\sqrt{7})(3 + \sqrt{7})$$

..... **[2]**

(b) Rationalise the denominator and simplify fully.

$$\frac{22}{5 - \sqrt{3}}$$

..... **[3]**

22 **(a)** Expand and simplify.

$(4 - 3\sqrt{2})(5 + 2\sqrt{2})$

..... **[2]**

(b) Rationalise the denominator and simplify.

$$\frac{11}{4 - \sqrt{5}}$$

..... **[3]**

(c) Solve the equation.

$x\sqrt{27} - 8 = x\sqrt{3} + 10$

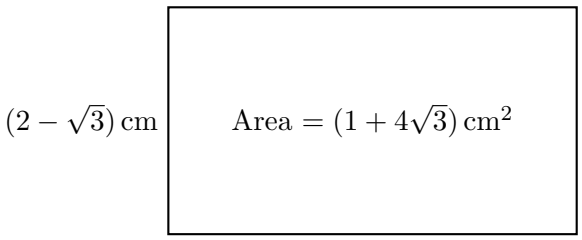
Give your answer in the form $k\sqrt{3}$ where k is an integer.

$x =$ ..... **[3]**

23 A rectangle has an area of $5 + \sqrt{7}$ cm^2 and a width of $3 - \sqrt{7}$ cm.
Calculate the length of the rectangle.
Give your answer in the form $a + b\sqrt{7}$ where a and b are integers.

..... *cm* **[4]**

24 The diagram shows a rectangle with width $(2 - \sqrt{3})$ cm and length $(a + b\sqrt{3})$ cm, where a and b are integers.
The area of the rectangle is $(1 + 4\sqrt{3})$ cm^2.



NOT TO SCALE

Work out the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[5]**

25 **(a)** Rationalise the denominator and simplify.

$$\frac{5+\sqrt{3}}{2-\sqrt{3}}$$

..... **[3]**

(b) Solve the equation.
Give your answer in the form $a+b\sqrt{2}$, where a and b are integers.

$$x(3-\sqrt{2})=14$$

$x =$ ..... **[3]**

26 **(a)** Express $\dfrac{5+\sqrt{3}}{2-\sqrt{3}}$ in the form $a+b\sqrt{3}$, where a and b are integers.

..... **[3]**

(b) A rectangle has an area of $12+4\sqrt{5}$ square centimetres and a width of $3+\sqrt{5}$ centimetres. Calculate the exact length of the rectangle, giving your answer in its simplest form.

Length = ... cm **[4]**

27 **(a)** Rationalise the denominator and simplify.

$$\frac{22}{5-\sqrt{3}}$$

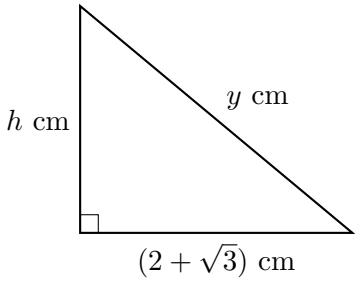
..... **[3]**

(b) Solve the equation.

$x\sqrt{2} = x + \sqrt{8}$ Give your answer in the form $a + b\sqrt{2}$, where a and b are integers.

$x =$ ..... **[4]**

28 A right-angled triangle has a base of $(2+\sqrt{3})$ cm and an area of $(5+3\sqrt{3})$ cm^2.



(a) Show that the exact height, h, is $(2+2\sqrt{3})$ cm.

[4]

(b) The hypotenuse of the triangle is y cm.
Work out the exact value of y^2.
Give your answer in the form $a+b\sqrt{3}$, where a and b are integers.

$y^2 =$ ..... **[3]**

29 **(a)** A rectangle has an area of $(5 + 2\sqrt{3})$ cm^2 and a length of $(2 + \sqrt{3})$ cm.

Work out the width of the rectangle in the form $a + b\sqrt{3}$, where a and b are integers.

..... cm **[3]**

(b) Solve the equation.

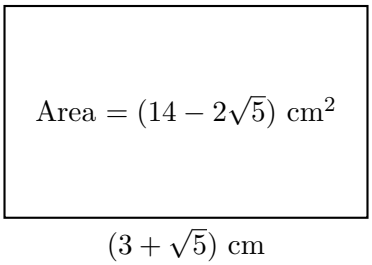
$$x\sqrt{5} - 2 = 2x + \sqrt{20}$$

Give your answer in the form $c + d\sqrt{5}$, where c and d are integers.

Show all your working.

$x =$ ..... **[4]**

30 A rectangle has length $(3+\sqrt{5})$ cm and area $(14-2\sqrt{5})$ cm^2.



(a) Calculate the width of the rectangle.
Give your answer in the form $a+b\sqrt{5}$, where a and b are integers.

..... cm **[4]**

(b) The length of the diagonal of the rectangle is d cm.
Show that $d^2 = 308 - 124\sqrt{5}$.

[3]

31 **(a)** Rationalise the denominator and simplify fully.

$$\frac{6}{3-\sqrt{3}}$$

..... **[4]**

(b) Write the following expression in the form $a\sqrt{2}+b\sqrt{3}$, where a and b are integers.

$$\frac{\sqrt{24}+\sqrt{54}}{\sqrt{3}}-\frac{10}{\sqrt{2}}$$

..... **[3]**

32 A right-angled triangle has a base of length $(3+\sqrt{2})$ cm and an area of $(5+2\sqrt{2})$ cm^2.

(a) Show that the height of the triangle is $\dfrac{2(5+2\sqrt{2})}{3+\sqrt{2}}$ cm.

[2]

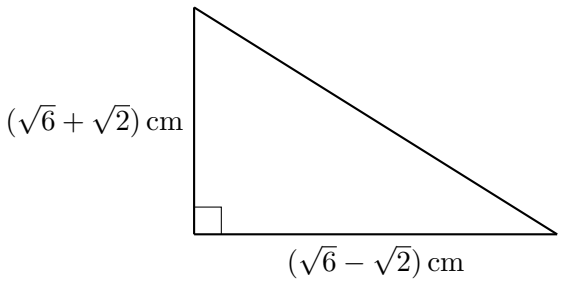
(b) Rationalise the denominator of $\dfrac{2(5+2\sqrt{2})}{3+\sqrt{2}}$ to express the height in the form $\dfrac{a+b\sqrt{2}}{c}$, where a, b and c are integers.

..... cm **[4]**

(c) Given instead that another expression for a length is $\dfrac{1}{\sqrt{7}-2}$, write this length in the form $\dfrac{\sqrt{7}+c}{d}$ where c and d are integers.

..... **[2]**

33 The diagram shows a right-angled triangle.



(a) Calculate the area of the triangle.

..... cm^2 **[3]**

(b) Show that the length of the hypotenuse is 4 cm.

[4]

(c) Another right-angled triangle is mathematically similar to this triangle.
The area of the similar triangle is 18 cm^2.
Calculate the length of its hypotenuse.

..... cm **[3]**

1 $5\sqrt{3} - 3\sqrt{3} + 2\sqrt{3}$ **[M1]**

$4\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 2]*

2 $5\sqrt{3}$ or $4\sqrt{3}$ or $3\sqrt{3}$ seen **[M1]**

$6\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 2]*

3 $6\sqrt{2}$ or $5\sqrt{2}$ or $3\sqrt{2}$ seen **[M1]**

$8\sqrt{2}$ **[A1]**

[Back to Question] *[Total Marks: 2]*

4 $5\sqrt{3}$ or $3\sqrt{3}$ or $2\sqrt{3}$ seen **[M1]**

$4\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 2]*

5 $\sqrt{25 \times 3} + \sqrt{16 \times 3} - \sqrt{9 \times 3} = 5\sqrt{3} + 4\sqrt{3} - 3\sqrt{3}$ **[M1]**

$6\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 2]*

6 **(a)** $3\sqrt{5}$ **[B1]**

(b) $2\sqrt{5}$ or $4\sqrt{5}$ seen **[M1]**

$6\sqrt{5}$ **[A1]**

[Back to Question] *[Total Marks: 3]*

7 $\sqrt{75} = 5\sqrt{3}$ **[B1]**

$\sqrt{27} = 3\sqrt{3}$ or $\sqrt{12} = 2\sqrt{3}$ **[B1]**

$6\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 3]*

8 $\sqrt{48} = 4\sqrt{3}$ **[B1]**

$\sqrt{12} = 2\sqrt{3}$ **[B1]**

$4\sqrt{3} - 2\sqrt{3} = 2\sqrt{3}$ **[B1]**

[Back to Question] *[Total Marks: 3]*

9 **(a)** $\sqrt{25 \times 3}$ or $\sqrt{25}\sqrt{3}$ **[M1]**

$5\sqrt{3}$ **[A1]**

(b) $6\sqrt{2} - 2\sqrt{2}$ **[M1]**

$4\sqrt{2}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

10 **(a)** **(i)** $3\sqrt{5} + 2\sqrt{5}$ **[M1]**

$5\sqrt{5}$ **[A1]**

(ii) Multiplies numerator and denominator by $\sqrt{3}$: $\frac{18\sqrt{3}}{3}$ **[M1]**

$6\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

11 **(a)** $3\sqrt{5}$ **[B1]**

(b) $6\sqrt{2} + 5\sqrt{2} - 3\sqrt{2}$ **[M1]**

$8\sqrt{2}$ **[A1]**

(c) $\frac{21\sqrt{7}}{\sqrt{7}\sqrt{7}}$ or $\frac{21\sqrt{7}}{7}$ **[M1]**

$3\sqrt{7}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

12 **(a)** $8 + 4\sqrt{3} - 2\sqrt{3} - 3$ **[M1]**

$5 + 2\sqrt{3}$ **[A1]**

(b) $\frac{21\sqrt{7}}{7}$ **[M1]**

$3\sqrt{7}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

13 **(a)** $9 - 3\sqrt{5} - 3\sqrt{5} + 5$ (or $9 - 6\sqrt{5} + 5$) **[M1]**

$14 - 6\sqrt{5}$ **[A1]**

(b) $\frac{14 \times \sqrt{7}}{\sqrt{7} \times \sqrt{7}}$ **[M1]**

$2\sqrt{7}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

14 **(a)** $12 + 3\sqrt{5} - 8\sqrt{5} - 10$ **[M1]**

$2 - 5\sqrt{5}$ **[A1]**

(b) Multiplies numerator and denominator by $\sqrt{7}$ to get $\frac{14\sqrt{7}}{7}$ **[M1]**

$2\sqrt{7}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

15 **(a)** **(a)** $12 + 3\sqrt{5} - 8\sqrt{5} - 2(5)$ or $12 - 5\sqrt{5} - 10$ **[M1]**

$2 - 5\sqrt{5}$ **[A1]**

(b) $\frac{15 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}}$ or $\frac{15\sqrt{3}}{3}$ **[M1]**

$5\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

16 **(a)** $12 - 3\sqrt{2} + 4\sqrt{2} - 2$ **[M1]**

$10 + \sqrt{2}$ **[A1]**

(b) Multiply numerator and denominator by $(5 + \sqrt{3})$ **[M1]**

Denominator evaluates to $25 - 3 = 22$ **[M1]**

$5 + \sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

17 **(a)** $20 - 9$ or $4(5) - 9$ **[M1]**

11 **[A1]**

(b) Multiplies numerator and denominator by $(3 + \sqrt{2})$ **[M1]**

Denominator of $9 - 2$ or 7 seen **[M1]**

$6 + 2\sqrt{2}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

18 **(a)** $12 + 15\sqrt{2} - 2\sqrt{2} - 5$ **[M1]**

$7 + 13\sqrt{2}$ **[A1]**

(b) Multiplies numerator and denominator by $(4 + \sqrt{3})$ **[M1]**

Denominator of $16 - 3$ or 13 **[M1]**

$8 + 2\sqrt{3}$ or $2(4 + \sqrt{3})$ **[A1]**

[Back to Question] *[Total Marks: 5]*

19 **(a)** $\frac{15\sqrt{5}}{5}$ **[M1]**

$3\sqrt{5}$ **[A1]**

(b) $10 - 2\sqrt{3} + 5\sqrt{3} - 3$ **[M1]**

$7 + 3\sqrt{3}$ **[A2]**

[Back to Question] *[Total Marks: 5]*

20 **(a)** $6 - 3\sqrt{5} + 2\sqrt{5} - 5$ **[M1]**

$1 - \sqrt{5}$ **[A1]**

(b) $\dfrac{14 \times \sqrt{7}}{\sqrt{7} \times \sqrt{7}}$ **[M1]**

$\dfrac{14\sqrt{7}}{7}$ **[M1]**

$2\sqrt{7}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

21 **(a)** $15 + 5\sqrt{7} - 6\sqrt{7} - 2(7)$ **[M1]**

$1 - \sqrt{7}$ **[A1]**

(b) Multiplies numerator and denominator by $(5 + \sqrt{3})$: $\dfrac{22(5 + \sqrt{3})}{(5 - \sqrt{3})(5 + \sqrt{3})}$ **[M1]**

Denominator becomes $25 - 3 = 22$ **[M1]**

$5 + \sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

22 **(a)** $20 + 8\sqrt{2} - 15\sqrt{2} - 6(2)$ **[M1]**

$8 - 7\sqrt{2}$ **[A1]**

(b) $\dfrac{11(4 + \sqrt{5})}{(4 - \sqrt{5})(4 + \sqrt{5})}$ **[M1]**

$\dfrac{11(4 + \sqrt{5})}{16 - 5}$ or $\dfrac{11(4 + \sqrt{5})}{11}$ **[M1]**

$4 + \sqrt{5}$ **[A1]**

(c) $3x\sqrt{3} - x\sqrt{3} = 18$ **[M1]**

$2x\sqrt{3} = 18$ or $x\sqrt{3} = 9$ **[M1]**

$x = \dfrac{9}{\sqrt{3}} = 3\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 8]*

23 $\times \dfrac{3 + \sqrt{7}}{3 + \sqrt{7}}$ **[M1]**

Numerator $15 + 5\sqrt{7} + 3\sqrt{7} + 7$ or $22 + 8\sqrt{7}$ **[B1]**

Denominator $9 - 7$ or 2 **[B1]**

$11 + 4\sqrt{7}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

24 Forms an equation for the area: $(a + b\sqrt{3})(2 - \sqrt{3}) = 1 + 4\sqrt{3}$ **[M1]**

Expands the brackets: $2a - a\sqrt{3} + 2b\sqrt{3} - 3b = 1 + 4\sqrt{3}$ **[M1]**

Equates rational and irrational parts to form simultaneous equations: $2a - 3b = 1$ and $-a + 2b = 4$ **[M1]**

Attempts to solve the simultaneous equations (e.g. multiplies second equation by 2: $-2a + 4b = 8$ and adds to first to get $b = 9$) **[M1]**

$a = 14$, $b = 9$ **[A1]**

[Back to Question] *[Total Marks: 5]*

25 (a) Multiplies numerator and denominator by $(2 + \sqrt{3})$ **[M1]**

Numerator expands to $10 + 5\sqrt{3} + 2\sqrt{3} + 3$ **[M1]**

Denominator is $4 - 3 = 1$ leading to final answer $13 + 7\sqrt{3}$ **[A1]**

(b) $x = \dfrac{14}{3 - \sqrt{2}}$ **[M1]**

Multiplies numerator and denominator by $(3 + \sqrt{2})$ to get $\dfrac{14(3 + \sqrt{2})}{9 - 2}$ **[M1]**

$x = 6 + 2\sqrt{2}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

26 (a) Multiply numerator and denominator by $2 + \sqrt{3}$ **[M1]**

Numerator becomes $(5 + \sqrt{3})(2 + \sqrt{3}) = 10 + 5\sqrt{3} + 2\sqrt{3} + 3$ (giving $13 + 7\sqrt{3}$) **[M1]**

$13 + 7\sqrt{3}$ **[A1]**

(b) Length $= \dfrac{12 + 4\sqrt{5}}{3 + \sqrt{5}}$ **[M1]**

Multiply numerator and denominator by $3 - \sqrt{5}$ **[M1]**

Numerator becomes $(12 + 4\sqrt{5})(3 - \sqrt{5}) = 36 - 12\sqrt{5} + 12\sqrt{5} - 20 = 16$ **[M1]**

4 **[A1]**

[Back to Question] *[Total Marks: 7]*

27 (a) Multiplies numerator and denominator by $(5 + \sqrt{3})$ **[M1]**

Denominator evaluates to $25 - 3$ or 22 **[M1]**

$5 + \sqrt{3}$ **[A1]**

(b) $x(\sqrt{2}-1)=\sqrt{8}$ or $x(\sqrt{2}-1)=2\sqrt{2}$ **[M1]**

$x=\dfrac{2\sqrt{2}}{\sqrt{2}-1}$ **[M1]**

Multiplies numerator and denominator by $(\sqrt{2}+1)$ **[M1]**

$4+2\sqrt{2}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

28 **(a)** $\dfrac{1}{2}\times(2+\sqrt{3})\times h=5+3\sqrt{3}$ **[M1]**

$h=\dfrac{10+6\sqrt{3}}{2+\sqrt{3}}$ **[M1]**

Multiply numerator and denominator by $(2-\sqrt{3})$ **[M1]**

$h=\dfrac{20-10\sqrt{3}+12\sqrt{3}-18}{4-3}=2+2\sqrt{3}$ **[A1]**

(b) $y^2=(2+\sqrt{3})^2+(2+2\sqrt{3})^2$ **[M1]**

$y^2=(4+4\sqrt{3}+3)+(4+8\sqrt{3}+12)$ **[M1]**

$23+12\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

29 **(a)** $\dfrac{5+2\sqrt{3}}{2+\sqrt{3}}$ **[M1]**

Multiplies numerator and denominator by $(2-\sqrt{3})$ **[M1]**

$4-\sqrt{3}$ **[A1]**

(b) $x\sqrt{5}-2x=2+\sqrt{20}$ or $x(\sqrt{5}-2)=\ldots$ **[M1]**

$x=\dfrac{2+2\sqrt{5}}{\sqrt{5}-2}$ **[M1]**

Multiplies numerator and denominator by $(\sqrt{5}+2)$ **[M1]**

$14+6\sqrt{5}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

30 **(a)** $\dfrac{14-2\sqrt{5}}{3+\sqrt{5}}$ **[M1]**

Multiplies numerator and denominator by $(3-\sqrt{5})$ **[M1]**

Numerator expands to $42-14\sqrt{5}-6\sqrt{5}+10$ (or $52-20\sqrt{5}$) **[M1]**

$13-5\sqrt{5}$ **[A1]**

(b) Uses Pythagoras' theorem: $(3+\sqrt{5})^2+(13-5\sqrt{5})^2$ **[M1]**

$14+6\sqrt{5}$ or $294-130\sqrt{5}$ seen **[M1]**

$308-124\sqrt{5}$ with no errors seen **[A1]**

[Back to Question] *[Total Marks: 7]*

31 **(a)** Multiplies numerator and denominator by $(3+\sqrt{3})$: $\dfrac{6(3+\sqrt{3})}{(3-\sqrt{3})(3+\sqrt{3})}$ **[M1]**

Expands denominator correctly: $9-3$ or 6 **[M1]**

$\dfrac{6(3+\sqrt{3})}{6}$ **[M1]**

$3+\sqrt{3}$ **[A1]**

(b) $\dfrac{2\sqrt{6}+3\sqrt{6}}{\sqrt{3}}$ or $\dfrac{5\sqrt{6}}{\sqrt{3}}$ or $5\sqrt{2}$ **[M1]**

$\dfrac{10\sqrt{2}}{2}$ or $5\sqrt{2}$ **[M1]**

0 (or $0\sqrt{2}+0\sqrt{3}$) **[A1]**

[Back to Question] *[Total Marks: 7]*

32 **(a)** Area $=\dfrac{1}{2}\times$ base $\times$ height $\implies 5+2\sqrt{2}=\dfrac{1}{2}\times(3+\sqrt{2})\times h$ **[M1]**

$h=\dfrac{2(5+2\sqrt{2})}{3+\sqrt{2}}$ **[A1]**

(b) Multiply numerator and denominator by $(3-\sqrt{2})$: $\dfrac{2(5+2\sqrt{2})(3-\sqrt{2})}{(3+\sqrt{2})(3-\sqrt{2})}$ **[M1]**

Denominator: $3^2-(\sqrt{2})^2=9-2=7$ **[B1]**

Numerator: $2(15-5\sqrt{2}+6\sqrt{2}-4)=2(11+\sqrt{2})=22+2\sqrt{2}$ **[M1]**

Height $=\dfrac{22+2\sqrt{2}}{7}$ **[A1]**

(c) $\dfrac{1(\sqrt{7}+2)}{(\sqrt{7}-2)(\sqrt{7}+2)}$ **[M1]**

$\dfrac{\sqrt{7}+2}{7-4}=\dfrac{\sqrt{7}+2}{3}$ **[A1]**

[Back to Question] *[Total Marks: 8]*

33 **(a)** $\dfrac{1}{2}\times(\sqrt{6}-\sqrt{2})(\sqrt{6}+\sqrt{2})$ **[M1]**

$\dfrac{1}{2}(6-2)$ **[M1]**

2 **[A1]**

(b) $(\text{Hypotenuse})^2=(\sqrt{6}-\sqrt{2})^2+(\sqrt{6}+\sqrt{2})^2$ **[M1]**

$(6-2\sqrt{12}+2)+(6+2\sqrt{12}+2)$ or $8-2\sqrt{12}+8+2\sqrt{12}$ **[M1]**

$(\text{Hypotenuse})^2=16$ **[A1]**

$\text{Hypotenuse}=\sqrt{16}=4$ **[A1]**

(c) Area scale factor $= \frac{18}{2} = 9$ **[M1]**

Length scale factor $= \sqrt{9} = 3$ **[M1]**

$4 \times 3 = 12$ **[A1]**

[Back to Question] *[Total Marks: 10]*